High Bandwidth Memory for AI

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Abstract

High-Bandwidth-Memory (HBM) technology is powering the accelerated computing and GPUs behind the AI revolution. This paper discusses the important role HBM architecture, performance, power, capacity, and RAS play in AI hardware. We will compare the HBM memory architecture with more traditional compute memory such as DDR5 focusing on Micron Technology's industry-leading HBM3E. We will end with a look towards the future technology innovations that will be essential to deliver continued HBM scaling.

Keywords: Bandwidth, Power, Memory, RAS, AI, DRAM, HBM, Heterogeneous Integration, TSV, 3D, 2.5D, Packaging

Introduction

The pace of innovation and adoption of Artificial Intelligence (AI) technologies has been unprecedented. At the core of this revolution are innovative new software models combined with the increasing computational capability of hardware designed to train and deploy AI models at scale. The exponential growth of AI—from large language models to real-time inference systems—has exposed fundamental limitations in traditional memory architectures. Roofline analysis reveals the severity of this bottleneck. The roofline model [Fig. 1] plots achievable performance (FLOP/s) against arithmetic intensity (FLOPs/byte), with "ceilings" representing hardware limits [1, 2].

HBM has emerged as a key enabler for AI systems, providing the bandwidth, capacity, and energy efficiency required to support GPUs [Table 1] [3, 4, 5]. GPUs require high memory bandwidth to efficiently handle the massive parallelism inherent in AI workloads, where Streaming Multiprocessors (SMs) execute thousands of threads concurrently. The high memory bandwidth ensures that data can be rapidly fetched and stored, minimizing latency and maximizing throughput during operations such as matrix multiplications, tensor computations, etc. [6]

Architecture

HBM is a heterogeneously integrated memory technology designed to deliver high bandwidth, capacity, and energy efficiency in a compact form factor. It utilizes a 3D stacked design with multiple DRAM die connected using through-silicon-vias (TSV), allowing for a highly parallel 3D interface.

A single HBM3E cube with 1K IO supports a 16X wider data interface compared to a DDR5 module. The wide interface supports accessing memory through 32 simultaneous and independent channels. This parallelism is achieved with the help of a high-speed TSV 3D PHY that interconnects all the DRAM die in the stack and a highly granular memory architecture within each DRAM die in the stack. HBM3E 24Gb DRAM die support 128 banks/die compared to 32 banks/die on a similar 24Gb DDR5 DRAM die. Memory banks are independent data arrays that can be accessed concurrently and organized as bank groups - higher number of banks allows greater parallelism in read/write operations. The increased bank count is supported by multiple parallel on-die data paths from the memory banks to the TSV PHY. HBM architecture is uniquely suited to serve AI workloads and a comparison with DDR5 can be seen in Fig 2. While HBM3E is one of the most energy efficient DRAM solutions, the cube still draws upwards of 20 Amps of current in a small 11mm x 11mm x 0.72mm form factor. Ensuring high quality and reliable operation requires careful vertical and horizontal power delivery network (PDN) design.

Advanced heterogeneous packaging technologies are then leveraged to make 100's of thousands of connections between every die within a cube at fine pitch (<25um) while also ensuring thermal conductivity is optimized to ensure the DRAM operating temperature stays within specifications even at peak operating conditions. The HBM cube is then integrated into a System-In-Package (SiP) along with host systems on an interposer [Fig. 3]. The short and dense channel routing on the interposer between host PHY and HBM PHY is a key enabler of high bandwidth at reduced power.

Due to the granular bank architecture and wide high-speed TSV 3D PHY, HBM DRAM die is significantly larger than a similar density DDR DRAM die [Fig. 4]. The larger silicon footprint combined with the advanced packaging and manufacturing flows involved result in approximately 3X more silicon being consumed for delivering HBM3E memory relative to similar DDR5 capacity.

Power & Performance

Micron Technology is proud of the HBM3E in production with industry-leading >1.2TB/s bandwidth with the world's lowest power consumption. HBM is one of the most cross-functionally complex products and required a highly co-optimized development cycle to achieve this innovative breakthrough. Design developed advanced low-power circuit design techniques, advanced signal and power integrity solutions. Technology Development developed Industry leading 1β DRAM and advanced CMOS and BEOL processes. Packaging developed advanced TSV and ubump processes with advanced die gap fill and design for thermal processes.

Reliability- Availability-Serviceability (RAS)

The memory-intense workloads running on large AI clusters for extended durations create unique RAS challenges and necessitate advanced RAS schemes on HBM. A paper on Llama3 training [7] highlighted the criticality of HBM in RAS performance of large AI clusters. Silent-Data-Corruption (SDC) was ranked the 10th most frequent fail pareto item. HBM3 has significantly improved RAS coverage with on-die symbol based ECC coverage [8]. While HBM products continue driving innovative solutions to achieve high level of RAS coverage at a cube level, the ultimate cluster level improvements will need close collaboration with system partners on achieving a holistic solution.

Future Work

As AI models, training, and inference workloads continue growing, demand for higher memory bandwidths and capacity at smaller power footprints will continue to drive innovation. The HBM base die is transitioning to advanced logic technology which will allow the opportunity to develop innovative new custom HBM solutions blurring the line between compute and memory, packaging technology is developing hybrid copper bonding and improving thermal conductivity, heterogeneous integrated systems continue driving towards further disaggregated 2.5D and 3D solutions.

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TABLE I. HBM Product Generations

Product	HBM1	HBM2	HBM2E	HBM3	HBM3E	HBM4
Timeline	2014	2018	2020	2022	2024	2026
# Channels	8	8	8	16	16	32
# Pseudo-Channels (PC)	-	16	16	32	32	64
PC Width (bits)	128	64	64	32	32	32
Burst Length	2	4	4	8	8	8
Prefetch (bits)	256	256	256	256	256	256
Data I/O's	1024	1024	1024	1024	1024	2048
Nominal Data Rate (Gbps)	1	2.4	3.6	6.4	8	6.4
Nominal Bandwidth (GB/s)	128	307	460	819	1024	1600
DRAM Density (Gb)	2	8	16	16	24	24
DRAM Stack Height	4	4/8	4/8	8/12	8+	8+
Cube Capacity (GB)	1GB	4/8GB	8/16GB	16/24GB	24GB+	24GB+

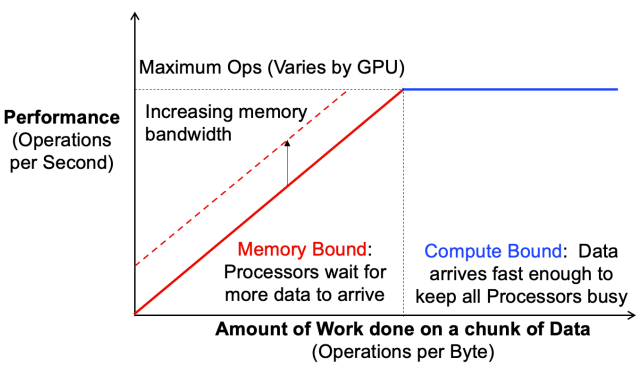

Fig. 1. Roofline Plot Analysis

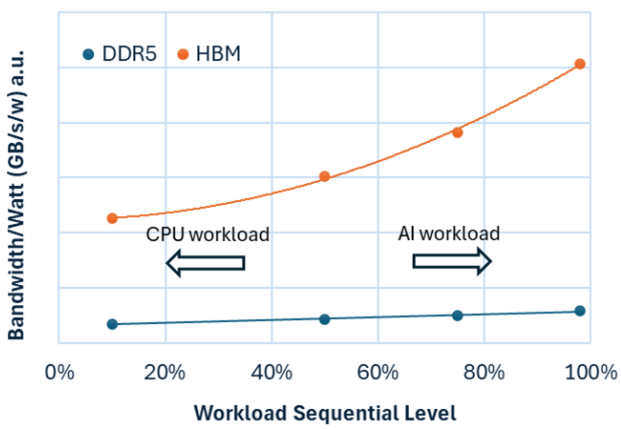

Fig. 2. Bandwidth per Watt performance comparison

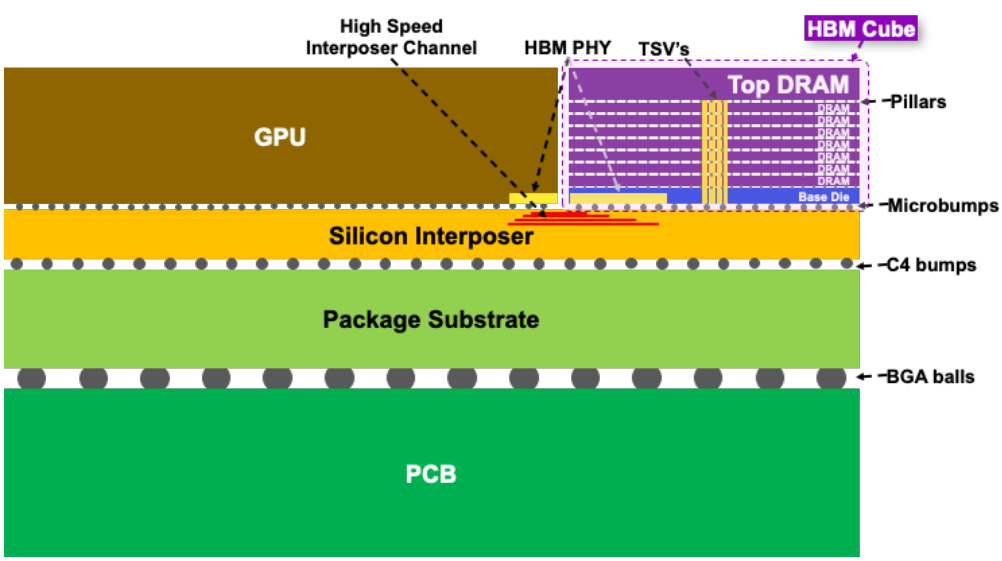

Fig. 3. System-In-Package (SiP) cross-sectional view

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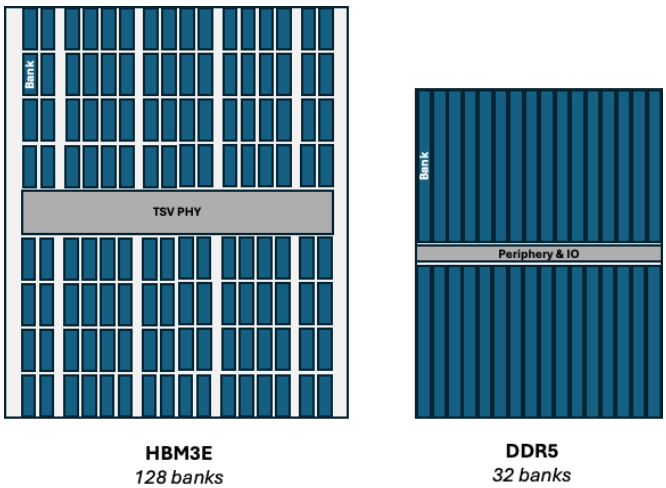

Fig. 4. HBM and DDR Bank Architecture

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